

Budget Proposal: Tahaffuz RAG System Upgrade

Date: June 8, 2026

Executive Summary

The Tahaffuz RAG (Retrieval-Augmented Generation) system currently experiences response times of 10-15 seconds due to Gemini free-tier limitations. This proposal outlines the infrastructure upgrade required to achieve:

- **Response times under 3 seconds**
- **Pashto and Sindhi language support** (spoken and written)
- **Improved accuracy and reliability**

Total Monthly Investment: \$150 USD (PKR 42,000)

Current System Limitations

Issue	Impact
Gemini Free Tier	Rate-limited, 10-15 second response times
No native Pashto/Sindhi	Users cannot interact in regional languages
SQLite FTS search	Keyword-based, misses semantic matches
No response caching	Repeated queries re-processed every time

Proposed Solution: Option B (Recommended)

1. Language Model (LLM)

Item	Provider	Purpose
GPT-4o-mini	OpenAI	Fast, accurate text generation

Specifications:

- Response time: 1-3 seconds
- Context window: 128K tokens
- Multilingual: Supports Urdu, Pashto, Sindhi, English
- Streaming: Yes (real-time responses)

Metric	Monthly Estimate
Estimated queries	10,000
Avg tokens per query	2,000
Monthly Cost	\$35 USD / PKR 9,800

2. Speech-to-Text (Voice Input)

Item	Provider	Purpose
Whisper API	OpenAI	Convert voice to text

Language Support:

- Urdu
- Pashto
- Sindhi
- English
- Auto-detection enabled

Metric	Monthly Estimate
Audio minutes	500 minutes
Rate	\$0.006/minute
Monthly Cost	\$20 USD / PKR 5,600

3. Text-to-Speech (Voice Output)

Item	Provider	Purpose
Azure Neural TTS	Microsoft	Natural voice responses

Voice Options:

- Urdu: Male & Female neural voices
- Pashto: Neural voice available
- Sindhi: Standard voice (custom training optional)
- English: Multiple accents

Metric	Monthly Estimate
Characters	1,000,000
Rate	\$16/1M characters
Monthly Cost	\$25 USD / PKR 7,000

4. Vector Database

Item	Provider	Purpose
Pinecone Starter	Pinecone	Semantic search

Benefits over SQLite FTS:

- True vector similarity search
- Sub-100ms query times
- Scales to millions of documents
- Managed infrastructure

Metric	Monthly Estimate
Vectors stored	Up to 100,000
Queries	Unlimited
Monthly Cost	\$0 USD (Free Tier)

5. Embeddings

Item	Provider	Purpose
text-embedding-3-small	OpenAI	Convert text to vectors

Specifications:

- Dimensions: 1536
- Multilingual support: Excellent
- Best-in-class for Urdu/Pashto/Sindhi

Metric	Monthly Estimate
Tokens embedded	5,000,000
Rate	\$0.02/1M tokens
Monthly Cost	\$10 USD / PKR 2,800

6. Caching Layer

Item	Provider	Purpose
Upstash Redis	Upstash	Cache frequent queries

Benefits:

- 80% faster repeat queries
- Reduces API costs
- Serverless (no management)

Metric	Monthly Estimate
Commands	100,000
Storage	256MB
Monthly Cost	\$10 USD / PKR 2,800

7. Contingency Buffer

Reserved for:

- Traffic spikes
- Additional testing

- Emergency scaling

Monthly Allocation	\$50 USD / PKR 14,000
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Cost Summary

Component	Provider	USD/Month	PKR/Month
LLM (GPT-4o-mini)	OpenAI	\$35	9,800
Speech-to-Text	OpenAI Whisper	\$20	5,600
Text-to-Speech	Azure Neural TTS	\$25	7,000
Vector Database	Pinecone	\$0	0
Embeddings	OpenAI	\$10	2,800
Caching	Upstash Redis	\$10	2,800
Contingency	-	\$50	14,000
TOTAL		\$150	PKR 42,000

Exchange rate used: 1 USD = 280 PKR

Annual Projection

Period	USD	PKR
Monthly	\$150	42,000
Quarterly	\$450	126,000
Annual	\$1,800	504,000

Expected Outcomes

Metric	Current	After Upgrade
Response Time	10-15 seconds	1-3 seconds
Languages Supported	3 (EN, UR, Dari)	5 (+Pashto, +Sindhi)
Voice Input	Not available	Full support
Voice Output	Not available	Full support
Search Accuracy	~70% (keyword)	~95% (semantic)
Uptime	Variable	99.9% SLA